

Original Contribution

The Role of Ozone as a Mediator of the Relationship Between Heat Waves and Mortality in 15 French Urban Areas

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Heat and tropospheric ozone have acute impacts on rates of premature death. Warm temperatures affect the photochemical processes in ozone formation, suggesting ozone as a mediator of the acute health effect of heat on mortality. We assembled a summertime daily time-series data set of 15 French urban areas during 2000–2015 to decompose the acute total effect of heat waves on mortality into natural direct and indirect effects using a regression-based product method under the potential outcomes framework. For each area, we estimated the effect of heat waves on mortality using a quasi-Poisson model with adjustment for covariates such as lagged nitrogen dioxide concentration, and we modeled ozone with a linear regression of heat waves and the same set of covariates. We pooled estimates across areas using random-effects models. We also provide R software code (R Foundation for Statistical Computing, Vienna, Austria) with which to reproduce or replicate our analysis. Most areas demonstrated evidence of mediation by ozone, with the pooled relative risks for natural indirect effects being 1.03 (95% confidence interval (CI): 1.02, 1.05), 1.03 (95% CI: 1.01, 1.04), and 1.04 (95% CI: 1.00, 1.07) for nonaccidental, cardiovascular, and respiratory mortality, respectively. We found evidence of a mediation effect by ozone in the association between heat waves and mortality in France which varied by geographic location and cause of mortality.

heat waves; mediation analysis; ozone

Abbreviations: CI, confidence interval; EEA, European Environment Agency; HHWS, heat-health warning systems; ICD-10, *International Classification of Diseases, Tenth Revision*; NDE, natural direct effect; NIE, natural indirect effect; RR, relative risk; TE, total effect.

Extreme temperature and ground-level ozone are major environmental determinants of population health, both driving increases in premature death. Short-term health effects of exposure to extreme heat have been largely described in the literature (1–6) and include cardiovascular and respiratory diseases. Similarly, short-term exposure to ozone, even at low concentrations, has been consistently shown to have negative health consequences, as it can cause a large range of cardiovascular and respiratory complications induced notably by oxidative stress, exacerbating existing health conditions and leading to premature death (7, 8).

Heat waves and high ozone concentrations frequently co-occur, as they are linked by complex atmospheric and chemical patterns. Specifically, warm temperatures influence the generation of tropospheric ozone, since hot, sunny

days can trigger photochemical processes (in conjunction with ozone precursors such as nitrogen oxides) that are at the basis of ozone formation. There has been debate in the literature about the role of ozone as a confounder, modifier, or mediator in the epidemiologic association between temperature and mortality (9, 10). Confounding bias is defined as the presence of a cause that is shared by an exposure *X* and an outcome *Y*, which results in an open backdoor path between *X* and *Y* (11); in this context, given that air pollution does not influence daily ambient temperature in the short term, there is no rationale for considering any air pollutant a true confounder when ambient temperature is the exposure of interest in short-term relation to a given health outcome. Alternatively, a number of recent studies have focused on the possible synergistic effect (or joint

effects) of high temperatures and ozone on population health (12–19). Indeed, simultaneous exposure to high temperature and ozone is likely to make individuals more susceptible to the effects of both environmental stressors; this is often considered through the inclusion of an interaction term or by considering different temperature strata in an exposure-response model (12–16). These studies considering ozone as an effect-measure modifier of heat-related health effects have denoted synergistic associations that often vary by region of study (12, 17–19).

Finally, ozone has also been proposed to be a causal intermediate factor in the relationship between temperature and mortality, since it is affected by temperature and can also affect mortality (9, 10). If ozone is part of the causal pathway between exposure and outcome, the total effect (TE) of temperature on mortality should be decomposed into an indirect effect, which operates through the ozone-effect component, and a direct effect, which consists of the temperature effect aside from the ozone-mediated effect (10). To the best of our knowledge, the role of ozone as a mediator in the relationship between heat and health has not been investigated, since no studies have assessed such a mechanism with empirical data. However, a better understanding of these dynamics would be particularly important for informing policy actions during a heat wave, as, in the context of climate change, heat-wave episodes are likely to occur more frequently and at higher intensity (20) and, concurrently, ozone concentrations are expected to increase in most urbanized areas (21–24); therefore, the potential health impact of this simultaneous exposure will become increasingly common.

During the last few decades, mitigation strategies that have been commonly adopted by various countries and cities worldwide have been based on 2 independent systems: heat-health warning systems (HHWS), which target heat-wave episodes as specific dangerous events for human health (25–28), and systems of air-quality alerts, which aim to prevent adverse health events associated with episodes of extreme air pollution (29–35). A deeper understanding of the relationship between high temperature, ozone, and mortality (and the variability across cities) is crucial to better adapt public-health policy measures implemented within the frameworks of these 2 systems.

By decomposing the effects of heat on mortality into direct and indirect effects through a causal mediation analysis, we can understand how ozone plays a role in the heat-mortality relationship and determine whether some urban areas would particularly benefit from specific actions to prevent ozone formation during heat waves. With this purpose, we provide a state-of-the-art causal mediation analysis to decompose the TE of heat waves on mortality in France during the summer periods from 2000 to 2015.

METHODS

Study population

Populations from 15 major urban areas in different regions of France were studied for the summer period, ranging from June 1 to September 30, for the years 2000–2015 (which

corresponds to the dates on which HHWS in France are active). In 2010, the sizes of urban populations ranged from 227,151 inhabitants in the Clermont-Ferrand area to 6.66 million inhabitants in Paris (36). The selected cities provide a representative panel of different climate profiles existing in France—namely the oceanic, temperate-oceanic, semicontinental, and Mediterranean climates (Figure 1). The urban areas studied and their main demographic, geographic, and climatic characteristics are listed in Table 1.

Outcome data

Daily mortality data for each urban area for the period 2000–2015 were obtained from the French National Institute of Health and Medical Research (Center of Epidemiology on the Medical Causes of Death). Daily mortality in the whole population was analyzed for the following causes: all nonaccidental mortality (*International Classification of Diseases, Tenth Revision* (ICD-10) codes A00–R99), cardiovascular causes (ICD-10 codes I00–I99), and respiratory causes (ICD-10 codes J00–J99).

Main definition of heat-wave exposure

Daily hourly minimal and maximal temperatures (in degrees Celsius) were provided by the French national meteorological service, Météo-France, for 1 reference station in each urban area, generally located at the airport (Table 2) (37). In order to identify heat waves for each urban area, we referred to the official method used by Météo-France to define heat waves, which is used to activate heat warning systems in France. Heat waves were defined as periods where the mean daily temperature exceeded the 97.5th percentile of the local (within each urban area) mean temperature distribution over the entire year (*initial threshold*) for at least 3 consecutive days, and for which at least 1 day had a recorded mean daily temperature above the 99.5th percentile of the local temperature distribution for the entire year (*peak threshold*) (37, 38). As sensitivity analyses, we also considered additional heat-wave definitions (see details below).

Air pollution data

We considered nitrogen dioxide a confounder of the relationship between ozone and mortality, since nitrogen dioxide level is one of the main precursors of tropospheric ozone formation. Two sources of air-quality data for the 2000–2015 period were utilized. First, daily ozone and nitrogen dioxide concentrations (in $\mu\text{g}/\text{m}^3$) were obtained from the open and publicly accessible database of the European Environment Agency (EEA) (39). The EEA's air-quality database contains multiannual time series of air-quality measurement data obtained by the regional entities in charge of monitoring air pollution in each participating country (including all European Union member states, as well as EEA-cooperating and other reporting countries), which are committed to transmitting their data to the agency. For all urban areas included in the study, relevant stations were selected by only considering stations which monitor pollutants at the hourly level.

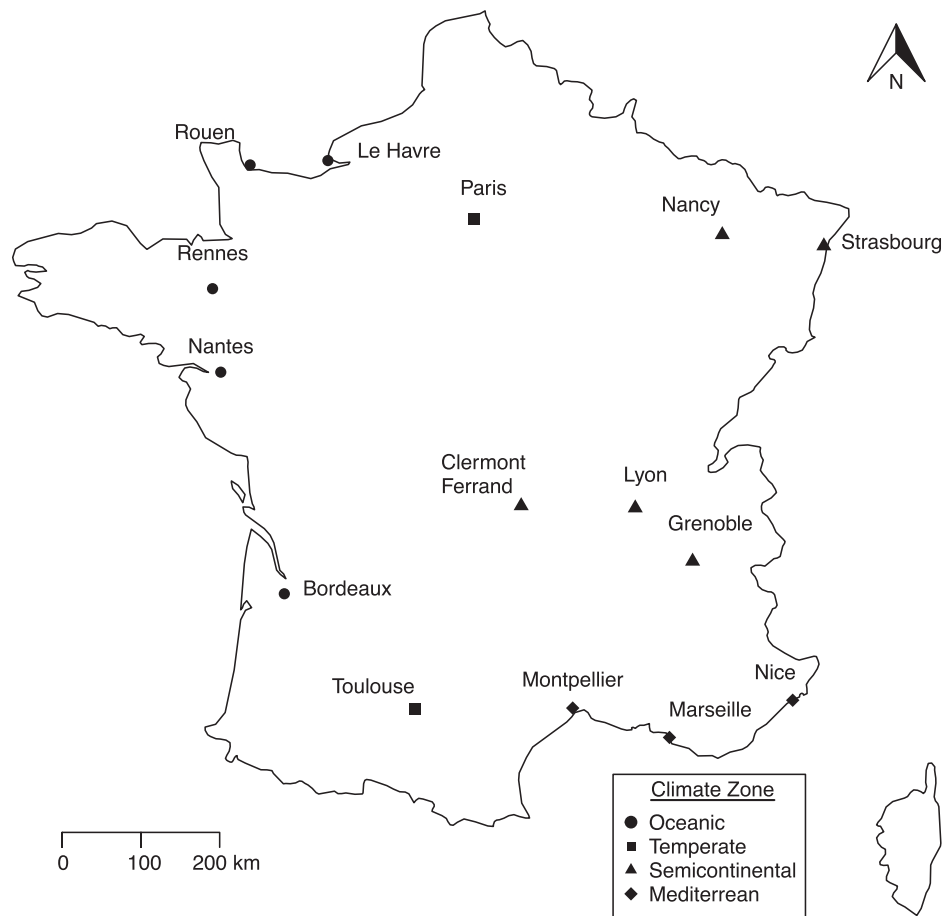


Figure 1. Urban areas included in a study of ozone as a mediator of the relationship between heat waves and mortality, by climate zone, France, 2000–2015.

Information on daily average concentrations for each urban area was then obtained by averaging hourly data from all of the stations within each urban area. The EEA's data were extracted using the R package *saqgetr* (40).

Second, data from the original source of the network of urban background monitors managed by local French air-quality surveillance organizations (<https://atmo-france.org/>) were also available for all of the urban areas included in the study, for the entire study period for ozone but only for the period 2007–2015 for nitrogen dioxide. Thus, these data were used for validation and for imputation of missing data.

Statistical analysis

Our analysis focused on the summer months, defined as the period from June 1 to September 30. We carried out a formal causal mediation analysis through a regression-based approach to understand the extent to which the association between heat waves (the exposure) and mortality (the outcome) was mediated by ozone (the mediator). This type of mediation analysis allows researchers to decompose the TE of heat waves on different causes of mortality into its natural

direct effect (NDE; the effect of the exposure to heat waves on mortality at a level of ozone concentrations observed during non-heat-wave events) and its natural indirect effect (NIE) operating through the mediator variable (the effect of the exposure to heat waves which operates through ozone).

Under the counterfactual (or potential outcomes) framework, the NDE is the contrast between mortality risk on heat-wave and non-heat-wave days holding ozone levels at the value they take on non-heat-wave days. The NIE, in contrast, compares the mortality risk present when ozone is at levels observed on heat-wave days versus non-heat-wave days, holding constant the exposure to a heat wave. In other words, the NIE represents the risk of death when exposure to heat waves is held constant and ozone exposure changes by a 1-unit increase to what it would have been for a change in the absence of a heat wave.

In order to estimate the NDE and the NIE, we fitted 2 multilevel regression models following the “product method” formalized by Baron and Kenny in 1986 (41) and adapted more recently under the potential outcomes framework (42–44). For each city, a quasi-Poisson regression (allowing for overdispersion) was first used for the outcome model,

Table 1. Demographic, Geographic, and Climatic Characteristics of 15 Urban Areas Included in a Study of Ozone as a Mediator of the Relationship Between Heat Waves and Mortality, France, 2000–2015

Climate Zone and Urban Area	Weather Station (Météo-France Name)	Location in France		Altitude, m	No. of Municipalities in Area	Population in 2010, no. of persons
		Degrees Longitude	Degrees Latitude			
Oceanic climate						
Bordeaux	Bordeaux Merignac ^a	−0.57	44	16	22	651,902
Le Havre	Le Havre—Cap de la Hève	0.13	49	70	16	241,037
Nantes	Nantes-Bougenais ^a	−1.55	47	20	27	601,460
Rennes	Rennes Saint-Jacques ^a	−1.68	48	35	4	240,769
Rouen	Rouen-Boos ^a	1.08	49	22	42	447,009
Temperate-oceanic climate						
Paris	Paris Montsouris	2.33	48	60	124	6,666,103
Toulouse	Toulouse Blagnac ^a	1.43	43	146	51	764,268
Semicontinental climate						
Clermont-Ferrand	Clermont-Ferrand	3.08	45	365	16	227,151
Grenoble	Grenoble ^a	5.72	45	212	46	472,741
Lyon	Lyon-Bron ^a	4.85	45	166	18	1,038,916
Nancy	Nancy-Essey ^a	6.20	48	222	38	331,903
Strasbourg	Strasbourg—Entzheim ^a	7.75	48	144	20	440,426
Mediterranean climate						
Marseille	Aéroport de Marignane ^a	5.40	43	20	8	970,751
Montpellier	Montpellier ^a	3.88	43	35	22	390,962
Nice	Nice ^a	7.25	43	10	4	435,428

^a Airport station.

which models daily mortality (Y) as a function of heat-wave episodes (hw), ozone concentration levels ($o3$), and a set of covariates c including day of the week, bank holidays, calendar month, and 2 temporal indicators: one for the day of the season (ranging from 1 on June 1 to 122 on September 30, with 4 degrees of freedom) and one for long-term trend (with 4 degrees of freedom) to address potential confounding by long-term trends on a time scale of approximately 4 years (45–48). Among the confounding factors which can influence both daily mortality and ozone formation (being one of its precursors), average nitrogen dioxide concentrations on the 2 previous days were also included in the set of covariates c .

$$E[Y|hw, o3, c] = \exp\left(\theta_0 + \theta_1 hw + \theta_2 o3 + \sum \theta_n c_n\right) \quad (1)$$

Then the mediator model was fitted through a linear regression linking ozone levels to heat-wave exposure and to the

same set of covariates.

$$E[o3|hw, c] = \beta_0 + \beta_1 hw + \sum \beta_n c_n \quad (2)$$

The NDE is given by the exponential of the coefficient θ_1 associated with the heat-wave variable (hw) in the outcome regression model:

$$NDE = \exp(\theta_1), \quad (3)$$

while the NIE is given by the exponential of the product of the coefficient θ_1 and the coefficient β_1 associated with the heat-wave variable (hw) in the mediator model:

$$NIE = \exp(\theta_1 \beta_1). \quad (4)$$

The TE will then be given by the product of these 2 components.

Table 2. Distributions of Temperature and Ozone Values¹ During the Summer Months^a in 15 Urban Areas, France, 2000–2015

Urban Area	No. of Observations	Temperature, °C				No. of Heat Waves	Ozone Concentration, µg/m ³		
		25th Percentile	Mean	75th Percentile	97.5th Percentile ^b	99.5th Percentile ^c	25th Percentile	Mean	75th Percentile
Bordeaux	1,927	17.95	20.29	22.30	25.34	28.17	52.44	65.02	76.18
Le Havre	1,943	15.26	16.97	18.20	21.00	23.99	51.40	60.67	68.09
Nantes	1,950	16.33	18.42	20.23	23.25	26.16	50.22	62.75	72.59
Rennes	1,909	5.90	17.93	19.75	22.35	25.30	42.07	54.10	62.82
Rouen	1,950	14.54	16.72	18.46	21.53	24.07	38.24	50.10	60.06
Paris	1,950	16.67	19.18	21.36	24.60	27.50	40.24	52.58	62.78
Toulouse	1,950	18.70	21.18	23.61	26.20	28.46	54.50	66.82	77.79
Clermont-Ferrand	1,944	16.35	19.00	21.50	24.70	27.84	53.17	65.23	76.07
Grenoble	1,935	16.30	18.76	21.35	24.05	26.30	42.21	57.83	71.98
Lyon	1,950	18.00	20.71	23.38	26.45	29.19	46.44	61.52	75.30
Nancy	1,948	15.70	18.20	20.76	23.60	26.30	38.84	52.01	62.28
Strasbourg	1,947	16.30	18.79	21.35	24.14	26.81	41.52	56.51	69.95
Marseille	1,948	21.18	23.25	25.32	27.35	28.89	59.62	70.45	80.98
Montpellier	1,937	20.70	22.62	24.60	26.84	28.34	62.50	73.17	83.70
Nice	1,948	21.25	22.96	24.70	26.50	28.04	66.13	77.64	89.11

^a Summer was defined as the period from June 1 to September 30.^b Initial threshold.^c Peak threshold.

Sensitivity analyses accounting for potential exposure-mediator interaction were performed through the 4-way decomposition method in order to verify whether the overall effect of the temperature on mortality can be decomposed into 4 components: the portion of the effect that is due to neither mediation nor interaction (the controlled direct effect), the portion due to just interaction (but not mediation, the reference interaction), the one due to both mediation and interaction (the mediated interaction), and the proportion due to just mediation (but not interaction, the pure indirect effect) (49). In the absence of an exposure-mediator interaction, like in the model presented above, the NDE coincides with the controlled direct effect, and they can be interpreted interchangeably.

When interpreting TEs, NIEs, and NDEs, we assumed no unmeasured confounding or mediator-outcome confounder affected by the exposure.

TEs, NDEs, and NIEs, expressed in terms of relative risks (RRs), and their 95% confidence intervals (CIs) were computed using bootstrapping procedures (700 replications). Urban area-specific estimates were then combined in a meta-analysis using random-effects models in order to obtain pooled estimates. The presence of heterogeneity was tested and reported using an I^2 statistic, which describes the percentage of variation across area-specific estimates that is due to heterogeneity rather than chance (50).

Finally, the proportion mediated (PM) was calculated for each urban area to quantify the contribution of ozone in the excess risk of mortality during heat-wave days, using the following formula:

$$PM = \frac{RR_{NDE} (RR_{NIE} - 1)}{(RR_{NDE} \times RR_{NIE} - 1)}. \quad (5)$$

We also conducted additional sensitivity analyses (details are provided in the Web Appendix, available at <https://doi.org/10.1093/aje/kwad032>) in which we 1) restricted heat-wave and non-heat-wave days with overlapping ozone values; 2) removed the year 2003 because of the major heat wave affecting France during that year; and 3) explored other heat-wave definitions.

All analysis were conducted using the R programming language, version 3.6.1 (R Foundation for Statistical Computing, Vienna, Austria) in the integrated development environment of RStudio (Posit PBC, Boston, Massachusetts), using the package *metafor* for meta-analyses and forest plots (51).

RESULTS

Descriptive statistics

During the study period, mean temperatures for the period from June to September varied considerably between urban areas, from 16.7°C (62.1°F) in the Rouen area to 23.3°C (73.9°F) in the Marseille area. The initial threshold values triggering the beginning of a heat wave and corresponding to the 97.5th percentile of the distribution of temperature values within each urban area and during the full year varied from a minimum of 21.0°C (69.8°F) in the Le Havre area to a maximum of 27.4°C (81.1°F) in the Marseille area

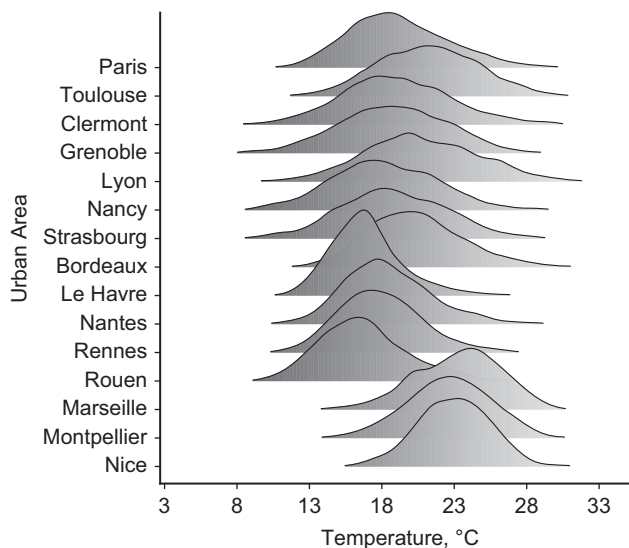


Figure 2. Distributions of temperatures in 15 urban areas during the summer months (June 1–September 30), France, 2000–2015. Areas are ranked according to their climate profile.

(Table 2). The highest value for the peak threshold (99.5th percentile) was reached in Lyon (29.2°C (84.5°F)), while the lowest corresponded again to Le Havre (24.0°C (75.2°F)). Complete information on the number of observed days (after selection of summer months and exclusion of days with missing data for mean temperature and ozone and nitrogen dioxide concentrations), temperature values, and the number of heat-wave days for each urban area are provided in Table 2. Temperature distributions for each urban area are also represented in Figure 2. Daily ozone concentrations varied from an average of 50.1 $\mu\text{g}/\text{m}^3$ in the Rouen area to 77.6 $\mu\text{g}/\text{m}^3$ in the Nice area. Ozone concentrations and distributions for each urban area are provided in Table 2 and Figure 3.

A total of 450,727 deaths from nonaccidental causes were recorded during summer periods from 2000 to 2015, across all of the urban areas considered; 117,047 of them were due to cardiovascular causes and 25,438 were due to respiratory causes. For each urban area, daily mortality was higher on days with a heat-wave episode than on normal days (Table 3 and Web Figures 1–3). In large urban centers such as Paris, daily mortality was 2-fold or almost 3-fold higher (for respiratory mortality) during heat waves as compared with other summer days (Table 3). In medium-sized and small areas (all of the areas considered except Paris, Lyon, and Marseille), the mean difference in daily mortality between heat-wave days and non-heat-wave days was 3 deaths for nonaccidental causes and 1 death for cardiovascular causes, while the mean difference for respiratory mortality was below 1.

Mediation analysis

Nonaccidental mortality. After adjustment for calendar variables and ozone and lagged nitrogen dioxide concentrations,

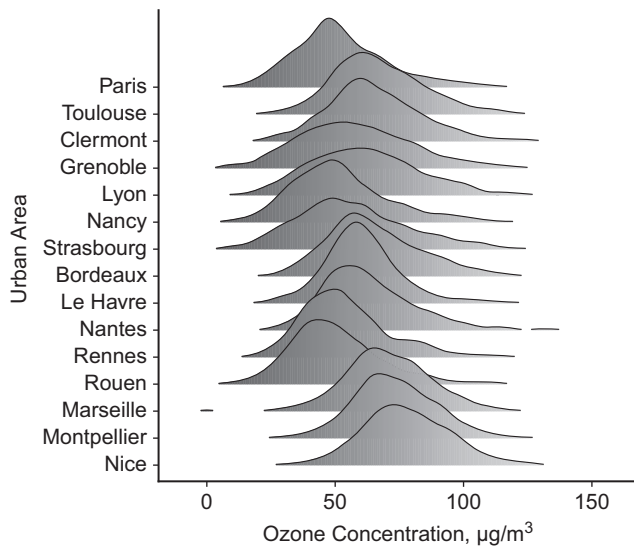


Figure 3. Distributions of ozone concentrations in 15 urban areas during the summer months (June 1–September 30), France, 2000–2015. Areas are ranked according to their climate profile.

associations between the occurrence of a heat wave and nonaccidental mortality were identified in all urban areas, with an RR_{TE} varying from 1.91 in Paris (95% CI: 1.38, 2.45) to 1.18 in Montpellier (95% CI: 0.99, 1.36) (Web Table 1 and Figure 4A). The estimated NDEs showed a stronger effect in the Paris ($RR_{NDE} = 1.81$, 95% CI: 1.31, 2.30) and Bordeaux ($RR_{NDE} = 1.57$, 95% CI: 1.33, 1.81) areas and a weaker effect, with a 95% CI overlapping the null value of 1, in the Montpellier area ($RR_{NDE} = 1.15$, 95% CI: 0.97, 1.34). As shown in Figure 4B, the pooled RR estimate for NDE among all urban areas was 1.33 (95% CI: 1.26, 1.40), with some heterogeneity across urban areas ($I^2 = 47.06\%$). A weak mediation effect was observed in almost all urban areas, except for Nancy, Bordeaux, and Le Havre (Figure 4C). Several RRs estimated for NIE showed large 95% CIs, particularly among areas with an oceanic climate profile (i.e., Le Havre, Nantes, Rennes, and Rouen). The pooled RR for NIE was estimated to be 1.03 (95% CI: 1.02, 1.05), with substantial heterogeneity across locations ($I^2 = 63.31\%$). The proportion of risk mediated by exposure to ozone for nonaccidental mortality varied from 31% in Nantes to 5% in Nice (Web Table 1).

Cardiovascular mortality. Results on cardiovascular mortality confirmed an effect of heat-wave exposure as well as a mediation effect of ozone for some urban areas (Web Table 2 and Figure 4). TE and NDE estimates were higher than those for nonaccidental mortality (e.g., for Clermont-Ferrand ($RR_{TE} = 1.89$, 95% CI: 1.45, 2.32) and Paris ($RR_{TE} = 1.99$, 95% CI: 1.38, 2.59)) but, for several urban areas, were more imprecise (Web Table 2; Figures 4D and Figure 4E). Consequently, the pooled estimate for NDE was higher than the one obtained for nonaccidental mortality ($RR_{NDE} = 1.38$, 95% CI: 1.26, 1.51) but with a larger 95% CI, which was due to the smaller number of cardiovascular

deaths. Some heterogeneity across areas was also observed for this estimate ($I^2 = 53.30\%$).

The estimates for the NIE on cardiovascular death were weak and not statistically different from 1 for almost all of the urban areas except Le Havre ($RR_{NIE} = 1.16$, 95% CI: 1.03, 1.28), Nantes ($RR_{NIE} = 1.11$, 95% CI: 1.02, 1.21), Paris ($RR_{NIE} = 1.06$, 95% CI: 1.01, 1.10), Toulouse ($RR_{NIE} = 1.07$, 95% CI: 1.02, 1.13), and Grenoble ($RR_{NIE} = 1.08$, 95% CI: 1.02, 1.15). Similarly to nonaccidental mortality, the pooled value for NIE for cardiovascular mortality was 1.03 (Figure 4F), with a 95% CI ranging from 1.01 to 1.04 and moderate heterogeneity across locations ($I^2 = 44.87\%$). The proportion of cardiovascular mortality risk mediated by ozone varied from 42% in Le Havre to 11% in Paris (Web Table 2).

Respiratory mortality. Associations between the occurrence of heat waves and respiratory mortality were detected in only some, more populated urban areas, namely the Paris ($RR_{TE} = 2.30$, 95% CI: 1.63, 2.98), Lyon ($RR_{TE} = 2.38$, 95% CI: 1.74, 3.03), Bordeaux ($RR_{TE} = 2.35$, 95% CI: 1.52, 3.18), Marseille ($RR_{TE} = 1.40$, 95% CI: 1.02, 1.77), and Toulouse ($RR_{TE} = 1.56$, 95% CI: 1.06, 2.06) areas, as well as in one smaller area, Strasbourg ($RR_{TE} = 2.09$, 95% CI: 1.45, 2.73) (Web Table 3 and Figure 4G). Estimates for NDEs were generally more imprecise for this cause of mortality than for the other two, and they were generally not statistically different from 1, except for the aforementioned areas (Figure 4H). The pooled RR for the NDE was estimated to be 1.67, with a 95% CI ranging from 1.47 to 1.86, and weak heterogeneity was observed across urban areas ($I^2 = 17.01\%$). Mediation effects were detected only in Nantes ($RR_{NIE} = 1.29$, 95% CI: 1.05, 1.52), Lyon ($RR_{NIE} = 1.16$, 95% CI: 1.07, 1.25), and Paris ($RR_{NIE} = 1.08$, 95% CI: 1.03, 1.13). The proportion of respiratory mortality risk mediated through ozone in these areas was 14% in Paris, 23% in Lyon, and 57% in Nantes. The pooled RR estimated for the NIE was 1.04 (95% CI: 1.00 to 1.07), with some heterogeneity across locations ($I^2 = 44.66\%$). We tested for the presence of a possible exposure-mediator interaction; no evidence of an interaction effect was found for any of the 3 causes of mortality. Descriptions of and results for sensitivity analyses are included in the Web Appendix, Web Tables 4–7, and Web Figures 4–10.

DISCUSSION

Our analysis showed that the effect of heat waves on mortality in France may be partially mediated by ozone, with important variability across urban areas. In some areas, like Nantes, mediation through ozone accounted for 31% of the total risk associated with heat-wave exposure for nonaccidental mortality, for 35% for cardiovascular mortality, and for up to 57% for respiratory mortality. In contrast, in a few urban areas like Nancy, evidence of a mediation effect of heat-wave exposure through ozone was not found for cardiovascular, respiratory, or nonaccidental mortality risk. Some heterogeneity was found across different locations studied in the decomposition of the TE of heat waves on nonaccidental

Table 3. Daily Mortality in 15 Urban Areas^a During the Summer Months^b on Normal Days and on Days With a Heat-Wave Episode, France, 2000–2015

Climate Zone and Urban Area	No. of Deaths per Day					
	Non-Heat-Wave Days			Heat-Wave Days		
	Minimum	Mean	Maximum	Minimum	Mean	Maximum
<i>Nonaccidental Causes of Death</i>						
Temperate-oceanic climate						
Paris	60	98.20	286	100	207.50	724
Toulouse	2	11.01	34	6	14.62	24
Semicontinental climate						
Clermont	0	5.17	15	4	8.04	16
Grenoble	0	7.23	19	3	9.47	16
Lyon	4	16.97	36	11	27.78	75
Nancy	0	6.29	16	0	9.31	24
Strasbourg	1	7.86	18	7	12.32	22
Oceanic climate						
Bordeaux	2	11.54	25	7	18.58	36
Le-Havre	0	5.28	14	2	6.96	15
Nantes	1	9.48	23	7	12.88	32
Rennes	0	3.57	12	2	5.03	9
Rouen	1	9.03	23	6	13.72	28
Mediterranean climate						
Marseille	7	20.09	39	12	24.89	40
Montpellier	0	5.83	17	2	6.93	14
Nice	2	11.14	24	4	15.27	33
<i>Cardiovascular Causes of Death</i>						
Temperate-oceanic climate						
Paris	8	23.58	56	13	50.48	179
Toulouse	0	2.97	11	0	4.24	9
Semicontinental climate						
Clermont	0	1.46	7	0	2.63	6
Grenoble	0	2.01	11	0	2.86	8
Lyon	0	4.43	14	2	7.09	22
Nancy	0	1.66	7	0	2.66	10
Strasbourg	0	2.21	9	0	3.45	7
Oceanic climate						
Bordeaux	0	3.26	13	3	5.81	12
Le-Havre	0	1.37	8	0	2.00	6
Nantes	0	2.58	11	0	3.54	10
Rennes	0	1.07	5	0	1.58	6
Rouen	0	2.49	12	0	3.40	8
Mediterranean climate						
Marseille	0	5.60	17	2	6.29	14
Montpellier	0	1.64	8	0	1.79	7
Nice	0	3.06	10	0	4.08	11

Table continues

Table 3. Continued

Climate Zone and Urban Area	No. of Deaths per Day					
	Non-Heat-Wave Days			Heat-Wave Days		
	Minimum	Mean	Maximum	Minimum	Mean	Maximum
<i>Respiratory Causes of Death</i>						
Temperate-oceanic climate						
Paris	0	5.45	20	2	14.52	52
Toulouse	0	0.56	5	0	0.90	3
Semicontinental climate						
Clermont	0	0.28	3	0	0.38	2
Grenoble	0	0.35	4	0	0.72	4
Lyon	0	0.89	5	0	2.41	7
Nancy	0	0.41	4	0	0.84	4
Strasbourg	0	0.44	6	0	0.95	3
Oceanic climate						
Bordeaux	0	0.66	5	0	1.65	6
Le-Havre	0	0.29	3	0	0.52	2
Nantes	0	0.51	6	0	1.04	4
Rennes	0	0.24	3	0	0.48	3
Rouen	0	0.50	5	0	0.96	3
Mediterranean climate						
Marseille	0	1.26	7	0	1.89	5
Montpellier	0	0.32	3	0	0.50	2
Nice	0	0.65	5	0	1.00	3

^a Ranked according to climate profile.

^b Summer was defined as the period from June 1 to September 30.

and cardiovascular mortality, while heterogeneity was less important when considering respiratory mortality.

These results are valuable for several reasons. First, from an etiological point of view, understanding the mechanisms driving the effect of high temperatures on population health through ozone has important implications for risk assessment; it allows us to better understand the excess mortality observed during heat waves. Both extreme temperatures and exposure to ozone concentrations have been shown to have adverse health effects, and several epidemiologic studies have confirmed that the mortality or morbidity risk associated with extreme temperatures is larger on days with elevated ozone concentrations (12, 15, 19, 52, 53). Our results regarding TEs are comparable to those of previous studies conducted in France that focused on the effects of extreme heat and mortality (1, 37, 45). Our mediation analysis contributes to a better understanding of this phenomenon by suggesting that, as a secondary pollutant that is generated by photochemical processes occurring on warm days, ozone can be considered a causal intermediate factor between heat waves and population health (10). Understanding where and when these mechanisms operate can help us determine the proportion of mortality/morbidity that is due to heat

exposure itself and which portion is attributable to ozone exposure.

Second, these results are particularly important not only for etiological reasons but also from the perspective of adapting public health interventions. National early warning systems and HHWS have been widely developed in the last few decades (28). HHWS use weather-based forecasting to predict the occurrence of a heat-wave event in order to alert stakeholders and trigger preventive measures to inform the general public and protect vulnerable populations. The definition of a heat wave varies by region and country according to geographic, population, and climatic characteristics, and it can be settled on the basis of different event-duration criteria and based on threshold values for temperature or on biometeorological index values that represent significant health effects (25–27). Even though evaluations of the effectiveness of HHWS are still scarce in the literature (25, 54), some authors have reported a reduction in the health impacts of heat waves following their implementation (55–57), while others have reported limited effects (54). Moreover, systems of air-quality alerts have been widely adopted in various big cities in different countries in order to prevent adverse health events associated with episodes of extreme air pollution

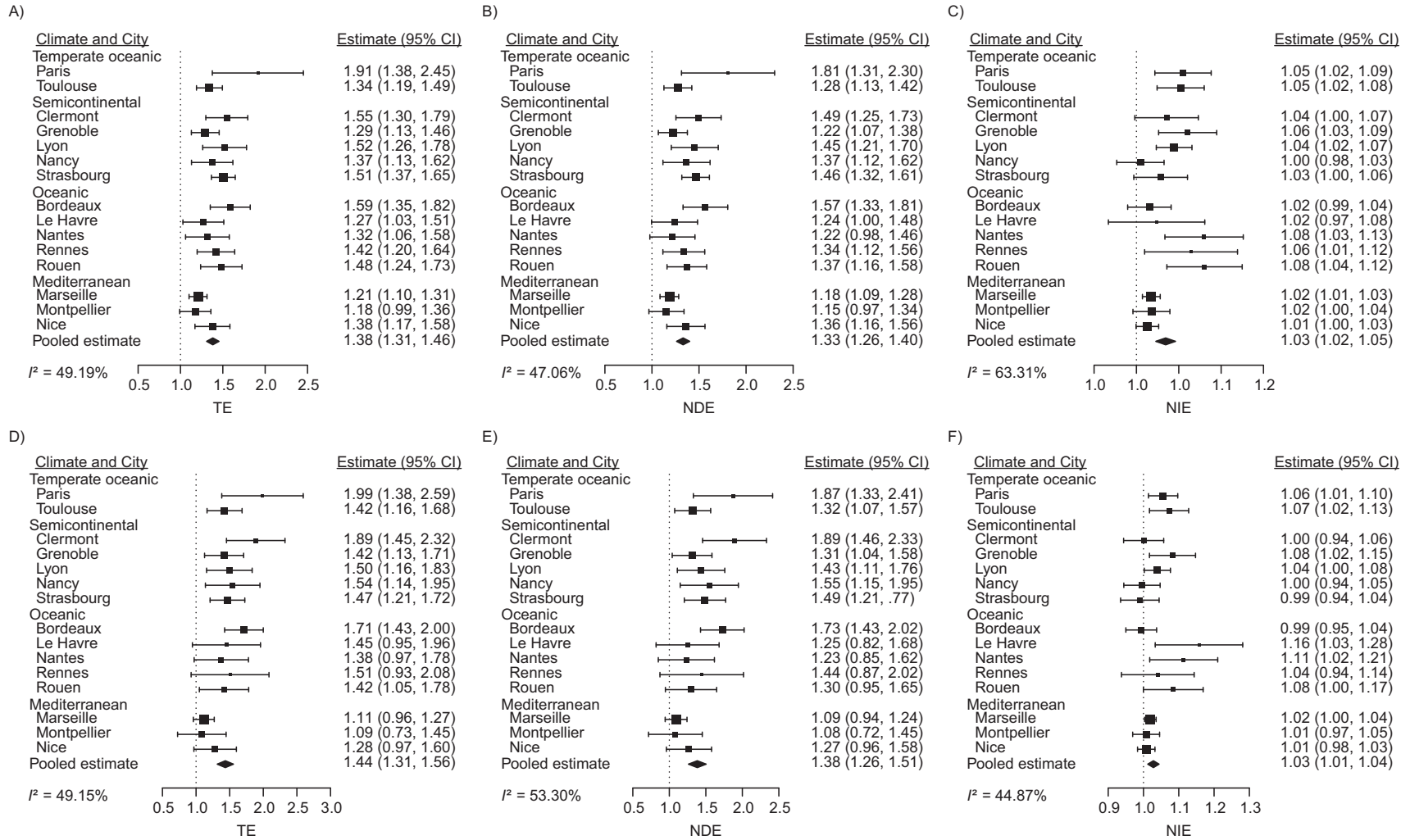


Figure Continues.

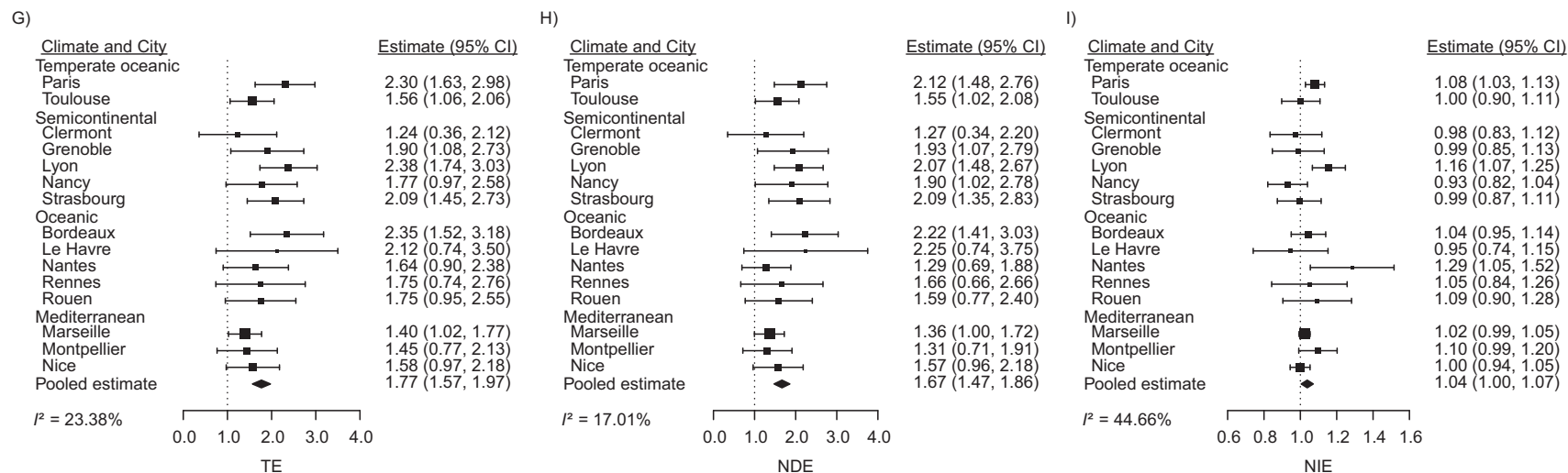


Figure 4. Urban-area (squares) and pooled (diamonds) relative risks of nonaccidental (first row), cardiovascular (second row), and respiratory (third row) mortality during the summer months (June 1–September 30), France, 2000–2015. Panels A, D, and G (left column) show estimates for the total effect (TE) of heat waves on nonaccidental, cardiovascular, and respiratory mortality, respectively. Panels B, E, and H (middle column) show estimates for the natural direct effect (NDE) of heat waves on mortality, and panels C, F, and I (right column) show estimates for the natural indirect effect (NIE) of heat waves on mortality. Bars, 95% confidence intervals (CIs).

(defined as “air pollution episodes”), which are identified through specific threshold values (58). These systems are focused on reducing emissions on days when air pollution concentrations exceed a given threshold, and despite their widespread acceptance, evidence about their impacts on population health is still limited (29–34). Preventive actions designed to reduce levels of ozone precursors (e.g., traffic-related emissions) for which a mediation effect is detected can help mitigate the TEs of heat waves on population health (59).

The results of our analyses showed some spatial variation among the cities studied. There are a few possible reasons for such heterogeneity. First, ozone formation during heat waves may vary across cities, depending on several factors. Indeed, ozone formation is based on complex photochemical processes and is directly linked to the presence of nitrogen dioxide and the oxidation of volatile organic compounds. For a given nitrogen dioxide concentration, ozone production rates can vary significantly depending on the presence of volatile organic compounds (60) and the level of sunlight, which varies considerably across cities in France. Other factors, like wind intensity and direction, can also drive tropospheric ozone concentrations in urban environments (61). Second, health effects of ozone exposure may vary across cities because of differences in several sociodemographic factors that have been shown to shape population vulnerability to air pollution. In particular, older individuals and communities with low socioeconomic status have been shown to be particularly susceptible to ozone (62–65).

Some limitations must be pointed out. First, we focused on the official climatological heat-wave definition, which corresponds to a small number of heat waves. This affected the statistical power of our analysis, especially for small urban areas and some specific causes of mortality. This may explain the lack of detection of a precise mediation effect on respiratory mortality in most of the urban areas studied. Second, we did not take into account a possible lag structure for heat-wave effects, and we did not consider the effect of a continuous variable for temperature, which would likely have a nonlinear relation with mortality. However, the choice of considering heat waves as a binary exposure (heat-wave and non-heat-wave days) is more pertinent when referring to the policy approach generally adopted in HHWS. Finally, our observations were made on the urban-area scale, and a finer spatial resolution would allow for more precise and reliable estimates; indeed, we only considered city-level exposures and had no access to information about individual exposures that people actually experience.

Some factors that could not be integrated into our analysis would be interesting to consider for future research: For example, the potential benefit on mortality of the French HHWS, started in 2004, could be considered and integrated into the analysis, as well as a possible difference in the health effects of heat waves according to their timing during the season, where a heat wave occurring at the beginning of summer may not have the same effect on population health as one occurring at the end of summer (66, 67).

In this study, using empirical data, we decomposed the TE of heat waves on mortality in 15 urban areas in France during summers from 2000 to 2015. Our results showed that

the effect of heat waves on mortality was partially mediated through ozone; this means that heat waves affect human health not only by exposing populations to extreme hot temperatures but also by generating higher ozone concentrations. To our knowledge, this is the first epidemiologic study to have empirically assessed which portion of heat-wave-related mortality is due to exposure to heat itself and which portion is attributable to ozone exposure. Further studies carried out in other countries with different climate profiles and population susceptibilities would be useful to better understand the role of ozone as a mediator in the relationship between heat waves and mortality.

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Data are available upon request from the corresponding author. Instructions for replicating all of the analysis are available on GitHub (https://github.com/aalari/Heat-wave_O3_Mediation).

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